

Internet Basics - Complete Easy Notes

Revision Notes with Real-Life Examples

1. What Happens When You Click a URL?

Flow:

URL → DNS Lookup → IP Address → TCP Handshake → HTTP Request → Server Processing → HTTP Response → Browser Rendering.

Pizza Shop Story:

1. You know the shop name (URL).
2. Ask for the address (DNS).
3. Travel to the address (IP Address).
4. Say Hello (TCP Handshake).
5. Order pizza (HTTP Request).
6. Chef prepares it (Server Processing).
7. Pizza is served (HTTP Response).
8. You enjoy it (Browser Rendering).

Remember: URL → DNS → TCP → Request → Response → Rendering.

2. DNS Lookup

DNS (Domain Name System) converts a domain name into an IP address.

Example: `www.google.com` → `142.xxx.xxx.xxx`.

Real-Life Example: Your phone converts 'Mom' into her phone number.

Interview Point: Humans remember names; computers communicate using IP addresses.

3. IP Address

Every device connected to the Internet has a unique address called an IP Address.

Public IP: Visible on the Internet (check using `WhatIsMyIPAddress`).

Private IP: Used only inside your home Wi-Fi (`192.168.x.x`).

Memory Trick: House Address = IP Address.

4. TCP 3-Way Handshake

Browser: Hello → Server: Hello, Ready → Browser: Let's Talk.

TCP provides reliable communication before data transfer starts.

Example: Making a phone call before talking.

5. HTTP Request

The browser requests a webpage from the server.

Example: `GET /index.html`.

Real-Life Example: Ordering food in a restaurant.

6. Server Processing

The server checks the request, accesses databases if needed, prepares the webpage, and creates the response.

Example: A chef cooking your ordered food.

7. HTTP Response

The server sends HTML, CSS, JavaScript, images, and videos back to the browser.

Example: Restaurant serving your meal.

8. Browser Rendering

The browser reads HTML (structure), CSS (design), and JavaScript (behavior) to display the website.

Example: Building a LEGO castle from separate pieces.

9. Router

A router forwards data packets between networks and chooses the best route.

Example: Google Maps choosing the fastest road.

10. Base64

Base64 converts binary data (images/files) into text so it can be sent in JSON, APIs, emails, etc.

Example: ...

Memory Trick: Image → Text → Image.

11. Internet Data Usage

ISPs count both Upload and Download.

Watching YouTube: small upload request + large video download = data usage.

12. File Sizes

8 bits = 1 Byte

1024 Bytes = 1 KB

1024 KB = 1 MB

1024 MB = 1 GB

1024 GB = 1 TB

1-Minute Interview Revision

- ★ DNS = Domain → IP Address
- ★ IP = Device Address
- ★ TCP = Reliable Connection
- ★ HTTP = Browser ↔ Server Communication
- ★ Server = Processes Request
- ★ Response = HTML/CSS/JS Files
- ★ Rendering = Display Webpage
- ★ Router = Best Path Selection
- ★ Base64 = Binary ↔ Text
- ★ Internet Usage = Upload + Download